Computational Astrophysics 2026-I
Project Homework #2

Magnetohydrodynamics — Simulation & Analysis

Objective

This project aims to deepen your understanding of magnetohydrodynamics (MHD) by reproducing and analyzing a physical problem using two complementary approaches: the **PLUTO code, which we saw in previous lectures** and a custom **Python implementation**. You will compare results, numerical errors, and provide a theoretical and state-of-the-art review of your chosen problem.

Instructions

1. Problem Selection

Choose **one** physical problem from the `PLUTO/Test_Problems/MHD` directory. The selected problem must be **at least 2D** (two-dimensional or higher).

2. Theoretical Framework & State of the Art

Your written review must include:

- A **theoretical framework** for your chosen problem, clearly stating the governing physical equations and boundary conditions.
- A **state-of-the-art review** summarizing recent literature and developments related to the problem.
- All relevant physical background needed to understand the simulation setup.

3. PLUTO Simulation

Select one example setup from within your chosen problem folder and reproduce it using the PLUTO code:

- Run the simulation to completion.
- Use **pyPLUTO** to generate plots of *all* physical variables (density, pressure, velocity components, magnetic field components, etc.).
- Ensure plots are clearly labeled and include colorbars, axis labels, and titles.

4. Python Reproduction

Independently reproduce the same physical problem using Python:

- Leverage functions and techniques covered in previous lectures (numerical solvers, plotting, data analysis, etc.).
- Implement the full pipeline: problem setup, solver, and post-processing.
- Plot all physical variables obtained from your Python solution with the same care as the PLUTO plots.

5. Comparative Analysis

Perform a rigorous comparison between the PLUTO and Python results:

- Produce **overlay plots** of corresponding physical variables from both approaches.
- Quantify and discuss **errors and discrepancies** (e.g., L_1/L_2 norms, relative differences).
- Analyze possible sources of error, numerical artifacts, convergence behavior, and physical insights.

Deliverables

#	Item	Description
1	**Report (PDF or Markdown)**	Theoretical framework, state-of-the-art review, problem description, methodology, plots, comparative analysis, and conclusions.
2	**Python Code**	Well-commented scripts used for the reproduction, solving, and analysis of the problem.
3	**PLUTO Input Files**	All configuration files and scripts used to run or post-process the PLUTO simulation.

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Evaluation Criteria

- **Theoretical Framework & Literature Review:** Completeness, accuracy, and clarity of the theoretical background and state-of-the-art summary.
- **PLUTO Simulation:** Correctness of the setup, reproducibility of results, and quality of pyPLUTO visualizations.
- **Python Implementation:** Code correctness, use of appropriate numerical methods, and overall quality of the implementation.
- **Comparative Analysis:** Depth of the error analysis, discussion of discrepancies, and physical interpretation of results.
- **Visualization:** Clarity, completeness, and professional presentation of all plots.
- **Report Quality:** Overall organization, writing clarity, and proper citation of all references.

> *Note* — **Academic Integrity:** Collaboration is encouraged for discussing concepts and approaches, but all code and written analysis must be your own original work. Cite all references and sources used in your report.